Sahaj Bhadja

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Summary

Junior Software Engineer with 2+ years building **C++/Java/Python systems**, **REST and MySQL integrations**, cross-platform applications, and **automated testing** across Windows and Linux environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant services** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load reliably.
- Achieved 99%+ reliability across 5K+ monthly jobs through **data-integration pipelines**, implementing validation, reconciliation, error handling, cloud storage, CI/CD releases, and stable production monitoring workflows reliably at enterprise scale.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production web platform** supporting 150+ instructors and 10K+ daily requests with React, Python APIs, AWS RDS, secure ingestion, and CI/CD.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during concurrent processing and sustained production traffic reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **React, Node.js, and Azure SQL platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with real-time updates across distributed business locations.
- Reduced resource wastage by 42% through **secure data synchronization**, implementing JWT access controls, WebSocket updates, automated tests, monitoring, and Docker/Jenkins deployments across eight modular product service domains.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Delivered **sub-200ms Flask APIs** across 50K+ booking records, implementing pricing, validation, payment webhooks, Firebase synchronization, and database transactions while preventing conflicting reservations across concurrent customer workflows reliably.
- Designed end-to-end system architecture across 12+ **backend integration points**, translating client requirements into API workflows and presenting technical decisions before implementation to align delivery expectations professionally and clearly.

Skills

Languages: C++, Java, Python, JavaScript, TypeScript, SQL, Bash
APIs & Applications: REST APIs, FastAPI, Flask, Spring Boot, Node.js, Express.js, React, WebSockets, Microservices
Data & Systems: MySQL, PostgreSQL, MongoDB, Azure SQL, AWS RDS, Firebase, Data Structures, Algorithms, Object-Oriented Programming, Design Patterns, Multithreading, Concurrent Programming
Platforms & Delivery: Windows, Linux, macOS, AWS, Azure, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD
Quality & Practices: Git, GitHub, Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Jest, Selenium, Debugging, Monitoring, Technical Documentation, Agile, Scrum, Requirements Analysis

Projects

FinNexus | Live
- Built a **TypeScript fintech platform** supporting 50K+ users and 1M+ transactions with secure bank linking, real-time synchronization, advanced filtering, Kubernetes deployment, and MongoDB-backed data access at production scale.

Low-Latency Market Data Gateway - Design Concept
- Designed a **low-latency market-data gateway** concept in C++ and Python with normalized message schemas, REST configuration APIs, MySQL persistence, validation, replay, structured logging, and deterministic integration tests.

CPU-Scheduling Simulator | Live
- Built a **C++ CPU-scheduling simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across scheduling scenarios.